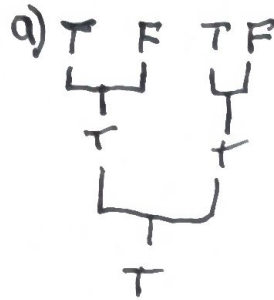


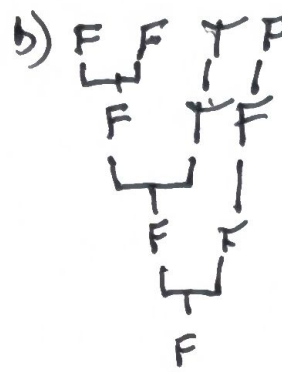
1. Given the following values for boolean variables p, q, r, s, evaluate each condition.

p = true, q = true, r = false, s = false

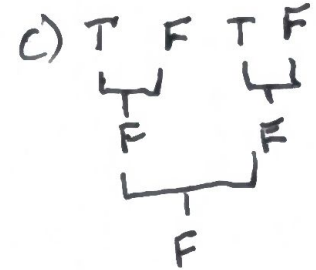
a.) (p || r) && (q || s)



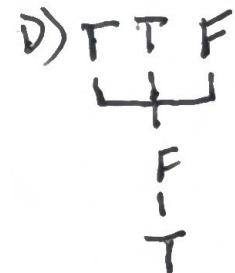
b.) !p || s && q || r



c.) (p && r) || (q && s)



d.) !(p && q && r)



2. Do a hand simulation of each program.

```

a.) public class HK4Q2a {
    public static void main(String args[]) {
        int alpha = 2;
        int beta = 5;

        if (alpha > beta) {
            alpha++;
        }

        if (alpha <= beta) {
            System.out.println(alpha);
        } else {
            System.out.println(beta);
        }

        System.out.println(alpha + " " + beta);
    }
}
  
```

ALPHA 2

BETA 5

SCREEN

2
2.5

```

b.) public class HK4Q2b {
    public static void main(String args[]) {
        int gamma = 3;
        int delta = 4;

        if (gamma == delta) {
            gamma--;
        } else if (gamma < delta) {
            gamma += 2;
            delta += 2;
        }

        System.out.println(delta + " " + gamma);
    }
}
  
```

GAMMA 3 5

DELTA 4 6

SCREEN

56